Karel Mundnich

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karel.mundnich@gmail.com • karel-mundnich • kmundnic.github.io

Summary

Applied Scientist III (Senior) at Amazon AGI with a Ph.D. in Electrical Engineering (USC). Technical lead for multi-stage LLM post-training recipes (SFT, DPO, RLVR, RLAIF), including recipes for 40+ models in the open-source SageMaker HyperPod Recipes, and builder of an evaluation framework covering 400+ public benchmarks. Before that, I spent three years building ASR and multimodal speech models at Amazon, including SpeechVerse, now a baseline in the Qwen2-Audio and Qwen2.5-Omni reports, with published work at ACL, EMNLP, Interspeech, and ICASSP.

Education

University of Southern California — Los Angeles, CA
Ph.D. in Electrical Engineering — *Aug '15 – May '21*
Thesis: Learning Multi-Annotator Subjective Label Embeddings
Advisor: Dr. Shri Narayanan

Universidad de Chile — Santiago, Chile
Electrical Engineering Diploma — *Mar '07 – Oct '13*
Thesis: Early Online Detection of High Volatility Clusters using Particle Filters
Advisor: Dr. Marcos Orchard
Honors: Graduated with highest honors
Note: Equivalent to M.Sc. + professional engineering certification

Relevant Experience

Amazon AGI — Seattle, WA
Applied Scientist III (Senior) — *May '26 – present*

- Technical lead for multi-stage LLM post-training recipes (combining SFT, DPO, and RL), extending our single-step fine-tuning techniques. Continue to oversee automation, evaluation pipelines, and onboarding of new models.

Amazon Web Services – AWS AI Labs — Seattle, WA
Applied Scientist III (Senior) — *Apr '25 – May '26*

- Technically led about 10 applied scientists, MLEs, and SDEs to launch recipes for post-training 40+ models with SFT, DPO, RLVR, and RLAIF in the open-source SageMaker HyperPod Recipes repository, covering Qwen, Llama, DeepSeek-R1 distilled, gpt-oss, Gemma, and Nemotron models.
- Built one evaluation wrapper around lm-evaluation-harness and Inspect AI that covers 400+ public benchmarks and also runs internal benchmarks, including DPO evaluation with an LLM judge. Used it, with automation pipelines, to onboard new models, and owned the end-to-end fine-tuning pipeline.

Applied Scientist II — *Jul '21 – Mar '25*

- *Speech LLMs and ASR* (Amazon Lex, Jul '21 – Oct '24):
  - Launched ASR models to production for Amazon Lex, including the first Catalan (ca-ES) and Portuguese (pt-BR and pt-PT) speech recognition models, and improved British English (en-GB) accuracy, with the largest gains on regional British accents.
  - Co-developed SpeechVerse, a multi-task audio-language model that beat task-specific baselines on 9 of 11 tasks and is now a baseline in the Qwen2-Audio and Qwen2.5-Omni technical reports [P1].
  - First-authored the extension to zero-resource speech translation and recognition (BLEU above 23 on unseen languages) [C2].
  - Contributed to adversarial robustness of speech LLMs [C3], and mentored an intern to a framework for RAG from speech without ASR [C1].
- *Embeddings and LLM fine-tuning* (Amazon Bedrock, Oct '24 – Mar '25):
  - Co-developed the next generation of Amazon Titan Text Embeddings models served in Bedrock, and created an LLM-based synthetic data generation pipeline to improve their training.

Selected Publications

Google Scholar (November 2026): 605 citations, h-index 11, i10-index 15.

Peer-reviewed conferences

[C1] Do June Min, Karel Mundnich, Andy Lapastora, Erfan Soltanmohammadi, Srikanth Ronanki, and Kyu Han. **Speech retrieval-augmented generation without automatic speech recognition**. In *ICASSP 2025 - 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025.

[C2] Karel Mundnich, Xing Niu, Prashant Mathur, Srikanth Ronanki, Brady Houston, Veera Raghavendra Elluru, Nilaksh Das, Zejiang Hou, Goeric Huybrechts, Anshu Bhatia, Daniel Garcia-Romero, Kyu J. Han, and Katrin Kirchhoff. **Zero-resource speech translation and recognition with LLMs**. In *ICASSP 2025 - 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025.

[C3] Raghuveer Peri, Sai Muralidhar Jayanthi, Srikanth Ronanki, Anshu Bhatia, Karel Mundnich, Saket Dingliwal, Nilaksh Das, Zejiang Hou, Goeric Huybrechts, Srikanth Vishnubhotla, Daniel Garcia-Romero, Sundararajan Srinivasan, Kyu Han, and Katrin Kirchhoff. **SpeechGuard: Exploring the adversarial robustness of multi-modal large language models**. In *Findings of the Association for Computational Linguistics ACL 2024*, pages 10018–10035, August 2024.

[C4] Tyler Vuong, Karel Mundnich, Dhanush Bekal, Veera Raghavendra Elluru, Srikanth Ronanki, and Sravan Bodapati. **AdaBERT-CTC: Leveraging BERT-CTC for text-only domain adaptation in ASR**. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pages 364–371, 2023.

[C5] Dhanush Bekal, Karthik Gopalakrishnan, Karel Mundnich, Srikanth Ronanki, Sravan Bodapati, and Katrin Kirchhoff. **A metric-driven approach to conformer layer pruning for efficient ASR inference**. In *Proceedings of Interspeech 2023*, pages 4079–4083, 2023.

[C6] Brandon M. Booth, Karel Mundnich, and Shrikanth Narayanan. **Fusing Annotations with Majority Vote Triplet Embeddings**. In *Proceedings of the 2018 on Audio/Visual Emotion Challenge and Workshop*, AVEC'18, pages 83–89. ACM, 2018. **Winner of the AVEC GES 2018 subchallenge**.

Journals

[J1] Karel Mundnich, Brandon M. Booth, Benjamin Girault, and Shrikanth Narayanan. **Generating Labels for Regression of Subjective Constructs using Triplet Embeddings**. *Pattern Recognition Letters*, 128:385–392, 2019.

Pre-prints

[P1] Nilaksh Das, Saket Dingliwal, Srikanth Ronanki, Rohit Paturi, Zhaocheng Huang, Prashant Mathur, Jie Yuan, Dhanush Bekal, Xing Niu, Sai Muralidhar Jayanthi, Xilai Li, Karel Mundnich, Monica Sunkara, Sundararajan Srinivasan, Kyu J. Han, and Katrin Kirchhoff. **SpeechVerse: A Large-scale Generalizable Audio Language Model**. In *arXiv:2405.08295v2*, 2024.

Skills & Background

Research Areas: LLM fine-tuning and post-training (SFT, DPO, RLVR, RLAIF), multimodal LLMs, automatic speech recognition, embedding models, retrieval-augmented generation (RAG), learning from subjective annotations.

Scientific Programming: Python, Julia.

Frameworks: PyTorch (FSDP), HuggingFace Transformers, verl, vLLM; familiar with Megatron-LM.

ML Systems: End-to-end fine-tuning pipelines, evaluation libraries, and automation for onboarding new models.

Languages: Spanish (*mother tongue*), English (*fluent*).

Awards & Distinctions

Patents: 3 patents filed at Amazon (pending).
AVEC Workshop 2018: Winner of the ACM AVEC GES sub-challenge 2018 [C6].